# Progressions

**1)** If a, b, c are in arithmetic progression, then the relation between a, b and c is:
**(a)** 2a = b + c **(b)** 2c = a + b **(c)** 3b = 2a + 3c **(d)** 2b = a + c
**2)** Find the 30th term of the AP: 10, 7, 4, ……
**(a)** 97 **(b)** 77 **(c)** -77 **(d)** -87
**3)** Which term of the AP: 3, 8, 13, 18, …… is 78?
**(a)** 12th **(b)** 14th **(c)** 15th **(d)** 16th
**4)** Find the number of terms in AP 7, 13, 19, …… 205
**(a)** 30 **(b)** 31 **(c)** 33 **(d)** 34
**5)** Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.
**(a)** 178 **(b)** 182 **(c)** 184 **(d)** 188
**6)** An AP consists of 50 terms of which 3rd term is 12 and the last term is 106. Find the 29th term.
**(a)** 62 **(b)** 63 **(c)** 64 **(d)** 65
**7)** How many terms of the series -9, -6, -3, …… must be taken so that the sum of all the terms is 45.
**(a)** 11 **(b)** 8 **(c)** 10 **(d)** 9
**8)** If the sum of first 14 terms of any AP is 1050 & its first term is 10, then find the 20th term.
**(a)** 200 **(b)** 201 **(c)** 203 **(d)** 202
**9)** What is the sum of all three-digit numbers which are divisible by 15?
**(a)** 41200 **(b)** 36825 **(c)** 32850 **(d)** 28750
**10)** What is the sum of all natural numbers between 100 and 400 which are divisible by 13?
**(a)** 5681 **(b)** 5334 **(c)** 5434 **(d)** 5761
**11)** How many 3 digit positive integers exist that when divided by 7 leave a remainder of 5?
**(a)** 128 **(b)** 142 **(c)** 143 **(d)** 141 **(e)** 129
**12)** Find the sum of all two-digit numbers which when divided by 5 leaves a remainder 2?
**(a)** 885 **(b)** 891 **(c)** 981 **(d)** 1001
**13)** What is the sum of all three-digit numbers that leave a remainder of '2' when divided by 3?
**(a)** 897 **(b)** 1,64,850 **(c)** 1,64,749 **(d)** 1,49,700
**14)** What is the sum of all positive integers up to 1000, which are divisible by 5 and are not divisible by 2?
**(a)** 10,050 **(b)** 5050 **(c)** 5000 **(d)** 50,000
**15)** Find the sum of first 32 terms of an AP if 5th term is 108 and 28th term is 250?
**(a)** 5728 **(b)** 5600 **(c)** 5278 **(d)** 5872
**16)** If the sum of first 14 terms of any AP is 1050 & its first term is 10, then find the 20th term.
**(a)** 200 **(b)** 201 **(c)** 203 **(d)** 202
**17)** Find the sum of 6 + 8 + 10 + 12 +14 …… +40.
**(a)** 400 **(b)** 1600 **(c)** 424 **(d)** 414
**18)** The sum of the three numbers in AP is 21 and the product of the first and third number of the sequence is 45. What are the three numbers?
**(a)** 5, 7, and 9 **(b)** 9, 7, and 5 **(c)** 3, 7, and 11 **(d)** Both (a) & (b)
**19)** If A= 1 - 10 + 3 - 12 + 5 - 14 + 7 + …… upto 60 terms, then what is the value of A?
**(a)** -360 **(b)** -310 **(c)** -240 **(d)** -270
**20)** Find the sum: 5 – 8 + 13 + 7 – 11 + 17 + 9 – 14 + 21 + …… upto 99 terms.
**(a)** 1843 **(b)** 1749 **(c)** 1815 **(d)** None of these
**21)** What is the value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \ldots\ldots + 12^2 - 11^2$.
**(a)** 5050 **(b)** 4985 **(c)** 4995 **(d)** 4950
**22)** Find the sum of all the common terms between the given series: [S1: 2, 9, 16, …… 632] & [S2: 7, 11, 15 …… 743]
**(a)** 6974 **(b)** 6750 **(c)** 6860 **(d)** 7140
**23)** If the square of 7th term of an AP with positive common difference equals the product of the 3rd and the 17th terms, then the ratio of the first term to the common difference is:
**(a)** 2:3 **(b)** 3:2 **(c)** 3:4 **(d)** 4:3
**24)** All the page numbers from a book are added, beginning from first page. However, one page number was added twice by mistake. The sum obtained was 1000. Which page number was added twice?
**(a)** 8 **(b)** 10 **(c)** 45 **(d)** 20
**25)** If the second and fifth term of an GP are 24 and 81, respectively, then find the GP.
**(a)** 16,24,36,54… **(b)** 8,12,16,24… **(c)** 18,24,36, 54… **(d)** None
**26)** If a rubber ball consistently bounces back 2/3 of the height from which it is dropped, what fraction of its original height will the ball bounce after being dropped and bounced four times without being stopped?
**(a)** 16/81 **(b)** 16/27 **(c)** 4/9 **(d)** 37/81 **(e)** 9/12
**27)** If a ball is thrown from a height of 500m on the ground, the ball bounces 4/5 times of its every last bounce. Find the total distance covered by the ball till it stops.
**(a)** 4500m **(b)** 5000m **(c)** 4200m **(d)** 4000m
**28)** The sum of third and ninth term of an AP is 8. Find the sum of the first 11 terms of the progression.
**(a)** 44 **(b)** 22 **(c)** 19 **(d)** None of the above
**29)** The sum of the fourth and twelfth term of an arithmetic progression is 20. What is the sum of the first 15 terms of the arithmetic progression?
**(a)** 300 **(b)** 120 **(c)** 150 **(d)** 170 **(e)** 270
**30)** If the first term is 216 and the common ratio is 5/6, what will be the 4th term of the GP?
**(a)** 129 **(b)** 123 **(c)** 127 **(d)** 125
**31)** The 1st term of a GP is 64 and the 5th term is 4. If the sum of all terms is 128, what is the common ratio?
**(a)** 1/8 **(b)** 2/5 **(c)** 1/2 **(d)** 1/4
**32)** If x, 2x+2, 3x+3 are first three terms of a GP then find its fourth term.
**(a)** -27/2 **(b)** -27 **(c)** 27 **(d)** None
**33)** Find the sum of $3+3^2+3^3+\ldots\ldots+3^8$
**(a)** 6561 **(b)** 6560 **(c)** 9840 **(d)** 3280

# Venn Diagrams

**1**) In a school, 50% students play cricket and 40% play football. If 10% of students play both the games, then what per cent of students play neither cricket nor football?
**(a)** 10% **(b)** 15% **(c)** 20% **(d)** 25%
**2)** In a group, 50 people speak Hindi, 20 speak Tamil and 10 speak both Hindi and Tamil, then number of people who speak Hindi or Tamil is:
**(a)** 60 **(b)** 70 **(c)** 80 **(d)** None of these
**3)** Out of the 35 students in a class, 13 students like vanilla and 15 students like chocolate. 10 students like neither of them. How many students like chocolate and vanilla?
**(a)** 25 **(b)** 2 **(c)** 3 **(d)** 13
**4)** In a group of students 70% can speak English and 65% can speak Hindi. If 27% of students can speak none of the two, then what percent of them can speak both the languages.
**(a)** 23% **(b)** 62% **(c)** 28% **(d)** 23%
**5)** At an overpriced department store there are total 112 customers. If 43 purchase shirts, 57 purchase pants and 38 purchase neither, how many purchased both shirts & pants?
**(a)** 26 **(b)** 74 **(c)** 14 **(d)** 38
**6)** In a school, 70 students are taking classes. 35 of them will be taking Accounting and 20 of them will be taking Economics. 7 of them are taking both of these classes. How many of the students are not in either class?
**(a)** 63 **(b)** 35 **(c)** 50 **(d)** 22
**7)** In a class of 100 students, 43 play basketball and 37 play baseball. 9 students play both. How many students do not play either sport?
**(a)** 38 **(b)** 71 **(c)** 29 **(d)** 20
**8)** Out of the 130 students who appeared for an examination, 52 students failed in Kannada, 72 students failed in Maths and 34 students failed in both Kannada and Maths. Then what is the total number of students who finally passed?
**(a)** 40 **(b)** 50 **(c)** 55 **(d)** 60
**9)** In a class of 60 students, 45 students like music, 50 students like dancing, 5 students like neither of them. Then the number of students who like both music and dancing is
**(a)** 35 **(b)** 40 **(c)** 50 **(d)** 55
**10)** In an examination 70% of the students passed in Maths, 60% of the students passed in English. 40% of the students passed in both subjects. How much percent of students failed in both the subjects?
**(a)** 10% **(b)** 15% **(c)** 30% **(d)** None of these
**11)** In an examination 35% students failed in economics, 45% failed in reasoning. If 20% students failed in both subjects. How much percent of students passed in both the subjects.
**(a)** 20% **(b)** 10% **(c)** 30% **(d)** 40%
**12)** In a population of cats, 10% are tabby colored, 5% are pregnant, and 3% are both tabby and pregnant. What percentage of cats are tabby but not pregnant?
**(a)** 7% **(b)** 10% **(c)** 3% **(d)** 7.5% **(e)** 5%
**13)** In a class of 160 students, each of them opts at least one language from among English, Hindi and Sanskrit. It is found that 130 students opt English, 120 students Hindi and 110 Sanskrit. If the students opt either only one language or all three languages, then what is the number of students who study all three languages?
**(a)** 40 **(b)** 60 **(c)** 80 **(d)** 100
**14)** In a city, there are 3 major newspapers A, B and C of which at least two are read by 35% of population. A and B are read by 15%. C is read by 45% and that all the three are read by 10%. Then the percentage of people who read the newspaper C alone is:
**(a)** 20 **(b)** 10 **(c)** 15 **(d)** 5
**15)** In a class test consisting of mathematics and physics, 80% passed in mathematics and 50% passed in physics. 15% failed in both the subjects. If 180 students passed in both the subjects, how many failed in both the subjects?
**(a)** 80 **(b)** 60 **(c)** 90 **(d)** 50
**16)** 30% people of India like cold weather, 40% like summer, 55% like autumn. 18% like cold and summer, 20% like summer and autumn, 12% like cold and autumn. 10% like all three weathers.
**(i)** How many like none of the weathers?
**(ii)** How many like only cold and summer?
**(iii)** How many like cold and autumn?
**(iv)** How many like at least one?
**(v)** How many like only one?
**(vi)** How many like at least two?
**17)** Data on the 468 students who took an examination on Chemistry, Physics and Mathematics, is as follows:
Passed in all the subjects-197
Failed in all the subjects-70
Failed in Chemistry-170
Failed in Physics-210
Failed in Mathematics-192
Passed in Chemistry only-64
Passed in Physics only-51
Passed in Mathematics only-46
**(i)** How many have failed in Chemistry only?
**(ii)** How many have failed in one subject only?
**(iii)** How many have passed in at least one subject?
**(iv)** How many have passed in at least two subjects?
**(v)** How many have passed in exactly two subjects?
**18)** One hundred twenty-five aliens descended on a set of film on extra-terrestrial beings. Of these, 40 had two noses, 30 had three legs, 20 had four ears, 10 had two noses and three legs, 12 had three legs and four ears, 5 had two noses and four ears, and 3 had all the three unusual features. How many were there without any of these unusual features?
**(a)** 5 **(b)** 35 **(c)** 80 **(d)** None of these